

Homework 4 — Solutions

Solution Key

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1. Normal mean, known variance: CRLB for unbiased estimators of μ

Assume $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$ with **known** σ^2 .

The log-likelihood is

$$\ell(\mu) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

The score and observed information are

$$U(\mu) = \frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu), \quad \frac{\partial^2 \ell}{\partial \mu^2} = -\frac{n}{\sigma^2}.$$

Hence the Fisher information for μ is

$$\mathcal{I}(\mu) = -\mathbb{E} \left[\frac{\partial^2 \ell}{\partial \mu^2} \right] = \frac{n}{\sigma^2}.$$

Therefore, for any unbiased estimator T of μ ,

$$\text{Var}(T) \geq \frac{1}{\mathcal{I}(\mu)} = \frac{\sigma^2}{n}.$$

2. Bernoulli(p): Fisher information and boundary behavior

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bernoulli}(p)$ and let $S = \sum_{i=1}^n X_i$.

(a) Derive $I(p)$

The likelihood and log-likelihood are

$$L(p) = p^S (1-p)^{n-S}, \quad \ell(p) = S \log p + (n-S) \log(1-p).$$

Differentiate:

$$U(p) = \frac{\partial \ell}{\partial p} = \frac{S}{p} - \frac{n-S}{1-p}.$$

Differentiate again:

$$\frac{\partial^2 \ell}{\partial p^2} = -\frac{S}{p^2} - \frac{n-S}{(1-p)^2}.$$

Thus the Fisher information is

$$I(p) = \mathcal{I}(p) = \mathbb{E} \left[-\frac{\partial^2 \ell}{\partial p^2} \right] = \mathbb{E} \left[\frac{S}{p^2} + \frac{n-S}{(1-p)^2} \right].$$

Using $\mathbb{E}[S] = np$ and $\mathbb{E}[n-S] = n(1-p)$,

$$I(p) = \frac{np}{p^2} + \frac{n(1-p)}{(1-p)^2} = \frac{n}{p} + \frac{n}{1-p} = \frac{n}{p(1-p)}.$$

(b) Show $I(p) \rightarrow \infty$ as $p \rightarrow 0$ or $p \rightarrow 1$

Since

$$I(p) = \frac{n}{p(1-p)},$$

we have $p(1-p) \rightarrow 0$ as $p \rightarrow 0$ or $p \rightarrow 1$, hence $I(p) \rightarrow \infty$ at either boundary.

3. Normal model (Question 1): variance of $\bar{X}$ and efficiency

(a) Compute $\text{Var}(\bar{X})$

Because $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $\text{Var}(X_i) = \sigma^2$,

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}.$$

(b) Compare to the CRLB

From Question 1, the CRLB for unbiased estimators of μ is σ^2/n .

(c) Conclude whether $\bar{X}$ is efficient

Since $\bar{X}$ is unbiased for μ and

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} = \text{CRLB},$$

$\bar{X}$ **attains** the CRLB and is therefore **efficient** (for estimating μ in this model).

4. Bernoulli model: $T = X_1$ as an estimator of p

(a) Is T unbiased?

Yes. Since $X_1 \sim \text{Bernoulli}(p)$,

$$\mathbb{E}[T] = \mathbb{E}[X_1] = p,$$

so T is unbiased for p .

(b) Compute $\text{Var}(T)$

$$\text{Var}(T) = \text{Var}(X_1) = p(1-p).$$

(c) Compare to the CRLB and comment on efficiency

From Question 2, the Fisher information is $I(p) = n/[p(1-p)]$, so the CRLB for unbiased estimators of p is

$$\text{Var}(\text{any unbiased } T) \geq \frac{1}{I(p)} = \frac{p(1-p)}{n}.$$

But for $T = X_1$,

$$\text{Var}(T) = p(1-p) = n \cdot \frac{p(1-p)}{n}.$$

So $T = X_1$ has variance n **times larger** than the CRLB (for $n > 1$) and is **not efficient** (except trivially when $n = 1$).

5. Exponential(λ): $I(\lambda)$, reparametrization, and transformation rule

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exponential}(\lambda)$ with density

$$f(x | \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0, \lambda > 0.$$

Let $S = \sum_{i=1}^n X_i$.

(a) Determine $I(\lambda)$

The log-likelihood is

$$\ell(\lambda) = n \log \lambda - \lambda S.$$

Differentiate:

$$U(\lambda) = \frac{\partial \ell}{\partial \lambda} = \frac{n}{\lambda} - S, \quad \frac{\partial^2 \ell}{\partial \lambda^2} = -\frac{n}{\lambda^2}.$$

Therefore,

$$I(\lambda) = -\mathbb{E} \left[\frac{\partial^2 \ell}{\partial \lambda^2} \right] = \frac{n}{\lambda^2}.$$

(b) Let $\theta = \log(\lambda)$. Compute $I(\theta)$

Here $\lambda = e^\theta$. Substitute into the log-likelihood:

$$\ell(\theta) = n\theta - e^\theta S.$$

Differentiate:

$$U(\theta) = \frac{\partial \ell}{\partial \theta} = n - e^\theta S, \quad \frac{\partial^2 \ell}{\partial \theta^2} = -e^\theta S.$$

Thus,

$$I(\theta) = \mathbb{E} \left[-\frac{\partial^2 \ell}{\partial \theta^2} \right] = \mathbb{E}[e^\theta S] = e^\theta \mathbb{E}[S].$$

Since $\mathbb{E}[X_i] = 1/\lambda = e^{-\theta}$, we have $\mathbb{E}[S] = ne^{-\theta}$. Hence

$$I(\theta) = e^\theta \cdot ne^{-\theta} = n.$$

(c) Verify the transformation rule

We have $\lambda = e^\theta$, so

$$\frac{d\lambda}{d\theta} = e^\theta = \lambda.$$

Using part (a),

$$I(\lambda) \left(\frac{d\lambda}{d\theta} \right)^2 = \frac{n}{\lambda^2} \cdot \lambda^2 = n = I(\theta),$$

which verifies the rule

$$I(\theta) = I(\lambda) \left(\frac{d\lambda}{d\theta} \right)^2.$$

6. Example where the CRLB exists but no estimator attains it

A standard example is estimating the **standard deviation** σ in a normal model with mean known to be 0:

$$X_1, \dots, X_n \stackrel{iid}{\sim} N(0, \sigma^2), \quad \sigma > 0.$$

The Fisher information for σ exists and is finite. The log-likelihood is

$$\ell(\sigma) = -n \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2 + \text{const},$$

so

$$\frac{\partial \ell}{\partial \sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n x_i^2, \quad \frac{\partial^2 \ell}{\partial \sigma^2} = \frac{n}{\sigma^2} - \frac{3}{\sigma^4} \sum_{i=1}^n x_i^2.$$

Using $\mathbb{E}[\sum x_i^2] = n\sigma^2$, we obtain

$$I(\sigma) = \mathbb{E}\left[-\frac{\partial^2 \ell}{\partial \sigma^2}\right] = -\left(\frac{n}{\sigma^2} - \frac{3}{\sigma^4} \mathbb{E}\left[\sum x_i^2\right]\right) = -\left(\frac{n}{\sigma^2} - \frac{3n\sigma^2}{\sigma^4}\right) = \frac{2n}{\sigma^2}.$$

So the CRLB (for unbiased estimators of σ) is

$$\text{Var}(T) \geq \frac{1}{I(\sigma)} = \frac{\sigma^2}{2n}.$$

However, **no estimator can attain this bound**. Equality in the CRLB requires (under regularity conditions) that

$$T(X) - \sigma = a(\sigma) U(\sigma),$$

where $U(\sigma) = \partial \ell / \partial \sigma$ is the score and $a(\sigma)$ is a nonrandom function of σ . But $U(\sigma)$ contains the unknown σ through terms like n/σ and $1/\sigma^3$, so any equality-form estimator would necessarily involve the unknown parameter σ (and thus cannot be a genuine statistic computable from the data alone). Therefore, although the CRLB exists, it is **not attainable** in this problem.